

Housewire Family Bonding Project Description

We built Housewire to turn a family home and the phones already in it into a cooperative escape room and game night system. We make each phone incomplete on its own: one person holds a clue, another controls the relevant action, and a third may need to carry or scan a device. Progress comes from talking, listening, delegating, and finishing together. AI prepares content and guidance behind the interface; it never competes with the family for attention.

The problem Housewire solves

Families may want shared time, but the default phone experience is private and parallel. Planning a game also asks one person to choose, explain, and run it. We reduce that setup work and change the phone from an individual destination into a prop within one shared event.

The family mechanism

We distribute complementary clues, permissions, and physical actions across devices. No screen contains the whole answer. A clue becomes useful only when its owner explains it, and the next step often belongs to someone else. These dependencies make clear communication part of the rules, give each player specific authority, and turn the finish into a shared result.

Three play modes

Mode	What the family does	How interdependence works
Escape Cases	Two to four phones become room stations. Families join by QR and use private clues, camera, audio, motion, haptics, and House Line.	We separate clue ownership, controls, and proof. The case advances only when players explain fragments, delegate movement, and complete shared checks.
Family Frequency	Two to four people pass one phone. Rounds mix choices, family knowledge, ordering, a 0 to 100 spectrum, and short text.	The answer owner seals the reference first. Semantic matching proposes a text score, and the owner may correct it. Exact reads can reward both people; in team play, bluffing only hurts the bluffer's side.
Circuit Race	Two crews race equivalent synchronized stages with private stations, team-specific hints, and one shared start clock.	Each crew relays riddles, reproduces rhythms, guides motion, and combines private gesture strips. The host validates the same underlying course for both teams.

The user journey

- 1 Choose** A family selects a case or mode. Judges can use solo rehearsal or Try vs ghost when only one phone is available.
- 2 Join** A live host opens the session; other phones scan the in-app QR or enter the same house code and relay address.
- 3 Check and assign** Each phone confirms available sensors or a touch fallback, then reveals only its private role.
- 4 Play and debrief** The host keeps synchronized state, verifies evidence, and releases the shared ending and run summary.

A live demonstration

The submitted 2 minute 53 second walkthrough starts on the Home screen, moves through the First Light tutorial, opens Escape Cases, configures Case Forge, generates a GPT-backed room, solves its first scene, and reveals the next. It then completes one Family Frequency round and previews Circuit Race with the contextual AI hint guide. The recording may appear slightly laggy because screen capture adds overhead; during normal play, the app runs smoothly. For a live judge demo, solo rehearsal keeps real answers, retries, hints, timing, validation, persistence, and debrief on one phone. With multiple phones, family members scan the in-app QR or enter the house code, receive private roles, and combine clues and actions to finish together.

How AI supports play

When a server-side key is configured, GPT-5.4 supports two bounded tasks. For Case Forge, it ranks locally compiled, already valid mechanical candidates and writes the story around the selected cut; local validators remain authoritative. For Family Frequency, it receives only style, count, and a random seed, then returns a safe question structure. Strict schemas and local checks reject malformed, unsafe, duplicate, or incomplete output. Any failure falls back silently to deterministic offline cases and question packs.

Separately, our on-device guide reads bounded game telemetry such as elapsed time, retries, stage progress, and sensor availability. It selects contextual non-spoiler hints and adjusts the level of support without changing a solution or completing proof. It does not listen to conversation or inspect camera frames. Housewire has no chatbot or visible AI persona.

Mobile and local first implementation

We built Housewire in React Native and TypeScript with Expo SDK 57 and Expo Router. Screens use clear providers and repositories for sensors, transport, generated cases, and persistence. Expo Go-safe adapters connect camera QR scanning, device motion and orientation, microphone and audio, and haptics. Permission-safe demo adapters and explicit touch actions preserve play when a capability is unavailable. Live sessions use host-authoritative state and the included local relay to synchronize phones on the same reachable network.

Privacy and trust

Core play needs no Housewire account. Family Frequency names and answers stay on the device. Camera content is used only during a chosen scan or clue and is not saved. Raw motion and microphone-level readings are processed locally and discarded; only bounded game evidence crosses the relay. House Line sends short recipient-scoped audio without transcription. We do not monitor family conversations.

Future work

- Online multiplayer will let relatives play together from different homes through secure invite links, private rooms, reconnectable sessions, and low-latency role audio while preserving local same-home play.
- A Family Case Studio will turn optional room photos, chosen themes, age ranges, and available devices into privacy-controlled escape rooms tailored to each household.
- New interaction packs will add spatial camera clues, directional sound, proximity, and coordinated sensor sequences, each paired with accessible touch, caption, and low-motion alternatives.
- Persistent family leagues will support saved teams, rematches, rotating hosts, shared case libraries, and asynchronous challenge windows for families whose schedules rarely overlap.